The electric displacement (Griffiths §4.3): structured notes

Griffiths, *Introduction to Electrodynamics* (5th ed.), pp. 181–186. Covers §4.3.1 and §4.3.3 (syllabus), with §4.3.2 and §4.3.4 as background. Equation numbers match the book. Figures are placeholder sketches for now.

Before you start

- Gauss's law in differential form: $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ (Section 2).
- Bound charges from §4.2: $\rho_b = -\nabla \cdot \mathbf{P}$ and $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ (Eqs. 4.11, 4.12).
- Electrostatic fields have zero curl, $\nabla \times \mathbf{E} = \mathbf{0}$ (Section 2).
- Maths: divergence, curl, the divergence theorem, Stokes' theorem.

1. Free and bound charge

- **Free charge** ρ_f: electrons on a conductor, ions in the material, or any charge not associated with $\mathbf{P}$.
- **Bound charge** ρ_b: the charge produced by polarization (§4.2).
- Total charge density inside a dielectric:

$$\rho = \rho_b + \rho_f \quad (4.20)$$

2. Derivation of the electric displacement (§4.3.1)

2.1 Gauss's law with both kinds of charge

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f$$

- Here $\mathbf{E}$ is the total field, not only the part due to polarization.

2.2 Combine the divergences

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f$$

> **DEFINITION**
> $\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P} \quad (4.21)$, the **electric displacement**.

> **RESULT**
> $\nabla \cdot \mathbf{D} = \rho_f \quad (4.22)$, and in integral form $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}} \quad (4.23)$.

2.3 Why this form is useful

- Equation 4.23 uses only the free charge, which is the quantity we control.

- Bound charge follows automatically once the free charge is in place, through the polarization it causes.
- In a typical problem ρ_f is known and ρ_b is not. Eq. 4.23 lets us start from what is known.
- Whenever the symmetry allows (spherical, cylindrical or plane), $\mathbf{D}$ follows directly from the usual Gauss's law method.

2.4 Surface bound charge

- Gauss's law in differential form cannot be applied exactly at the surface, because ρ_b is infinite there (the polarization drops abruptly to zero). This is the surface term σ_b (footnote 7).
- If the edge is given a thin transition zone where $\mathbf{P}$ tapers to zero, ρ_b is large but finite, and Gauss's law holds everywhere. The integral form (4.23) avoids the issue altogether.

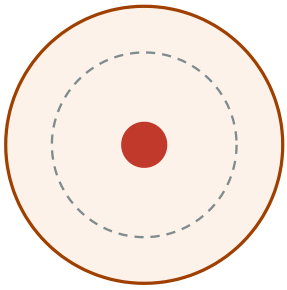
3. Worked example: wire with rubber insulation (Example 4.4)

- Uniform line charge λ on a long wire, surrounded by insulation out to radius a.
- Gaussian cylinder of radius s and length L. By symmetry $\mathbf{D}$ is radial and constant on the surface. The free charge inside is λL.

$$D\,(2\pi s L) = \lambda L$$

> **RESULT**
> $\mathbf{D} = \dfrac{\lambda}{2\pi s}\,\hat{\mathbf{s}}$ (4.24), valid both inside and outside the insulation.

- Outside ($s > a$): $\mathbf{P} = 0$, so $\mathbf{E} = \mathbf{D}/\epsilon_0 = \dfrac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}}$.
- Inside: $\mathbf{E}$ cannot be found from $\mathbf{D}$ alone, because $\mathbf{P}$ is not yet known. This needs the material's response, given in §4.4.



Placeholder: wire, insulation, Gaussian cylinder

Figure 1. Placeholder sketch for Example 4.4. To be replaced by the Codex version.

4. Method: finding D and then E

1. List the free charges only.
2. Check the symmetry. If it is spherical, cylindrical or planar, continue; otherwise do not assume $\mathbf{D}$ is set by free charge alone (§5).
3. Choose a Gaussian surface matching the symmetry and apply Eq. 4.23.
4. Find $\mathbf{E}$ from $\mathbf{E} = (\mathbf{D} - \mathbf{P})/\epsilon_0$ (Eq. 4.21) once $\mathbf{P}$ is known.

5. Pitfalls (§4.3.2: beyond syllabus)

- Eq. 4.22 looks like Gauss's law with ρ_f in place of ρ, so it is tempting to say "$\mathbf{D}$ is just $\mathbf{E}$ with free charge as the source". This is false: there is no Coulomb's law for $\mathbf{D}$.
- A divergence alone does not fix a vector field; the curl is needed too.
- The curl of $\mathbf{D}$ is not zero in general:

$$\nabla \times \mathbf{D} = \epsilon_0(\nabla \times \mathbf{E}) + \nabla \times \mathbf{P} = \nabla \times \mathbf{P} \quad (4.25)$$

- So $\mathbf{D}$ cannot be written as the gradient of a scalar: there is no potential for $\mathbf{D}$.
- Example: a bar electret with no free charge. If $\mathbf{D}$ were set by free charge alone, $\mathbf{D} = 0$ everywhere and $\mathbf{E} = -\mathbf{P}/\epsilon_0$ inside. That is wrong, so the curl term must be kept.
- Advice from the book: use symmetry first. With symmetry, $\nabla \times \mathbf{P}$ is automatically zero, so Gauss's law gives $\mathbf{D}$ directly.

6. Boundary conditions (§4.3.3: in syllabus)

6.1 Normal component of D

- Pillbox across the interface, with area A, height tending to zero. Only the two flat faces contribute to $\oint \mathbf{D} \cdot d\mathbf{a}$.
- Free charge on the interface $\sigma_f A$ is enclosed, so

> **RESULT**
> $$D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f \quad (4.26)$$

6.2 Tangential component of D

- Take a thin rectangular loop across the interface and apply Stokes' theorem to Eq. 4.25. The field $\nabla \times \mathbf{P}$ is integrated over the loop's area, which shrinks to zero, but $\mathbf{P}$ itself is finite. The loop integral of $\mathbf{P}$ gives the tangential difference times the loop length.

> **RESULT**
> $$\mathbf{D}^{\parallel}_{\text{above}} - \mathbf{D}^{\parallel}_{\text{below}} = \mathbf{P}^{\parallel}_{\text{above}} - \mathbf{P}^{\parallel}_{\text{below}} \quad (4.27)$$

6.3 Boundary conditions on E

- Written in terms of total charge, the normal and tangential conditions on $\mathbf{E}$ are (Eqs. 2.31, 2.32):

$$E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} = \frac{\sigma}{\epsilon_0} \quad (4.28)$$

$$\mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = 0 \quad (4.29)$$

- Use (4.26)–(4.27) when $\mathbf{P}$ is known and free charge is the given data. Use (4.28)–(4.29) for the field directly. Probs. 4.16 and 4.17 are suggested.

7. Ionic crystals: the ambiguity (§4.3.4: background)

- In a salt crystal, Na^+ and Cl^- can be paired into dipoles in several ways. Each choice gives a different $\mathbf{P}$ (up, down, left, right), and a different split of charge into free and bound parts.
- So the separation into $\mathbf{P}$, free charge and bound charge is not uniquely defined for crystals.
- What is physical is the total charge density, which is zero in the salt crystal, and changes in $\mathbf{P}$ when a field is switched on.

8. What you should be able to do

- Derive Eq. 4.22 from Gauss's law with $\rho = \rho_b + \rho_f$.
- Find $\mathbf{D}$ for a symmetric free charge distribution using Eq. 4.23 (Example 4.4).
- State and derive Eqs. 4.26 and 4.27 with a pillbox and a thin loop.
- Explain why $\mathbf{D}$ cannot be found from free charge alone when $\nabla \times \mathbf{P} \neq 0$.

9. Problems in the excerpt (not solved here)

- Prob. 4.15: thick spherical shell with $\mathbf{P} = (k/r)\hat{\mathbf{r}}$. Find E in three regions by two methods.
- Prob. 4.16: field at the centre of spherical, needle and wafer cavities in a polarized dielectric.
- Prob. 4.17: sketches of P, E and D for the bar electret (Prob. 4.11), with $L \approx 2a$.

10. Open points

- Sign convention of the normal in Eqs. 4.26 and 4.28 to be checked against Section 2.3.5.
- Footnote 7 (delta-function derivative) is stated, not derived here.
- Figures 4.17–4.22 are placeholders in these notes.
- Section 4.4 (linear dielectrics, $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$) is the next section, not included here.